Monitoring the Water Extents of the Garda Lake

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Abstract

This project aims to compare the extents of Garda lakes during the period over spring and summer of 2022 and 2023, using Sentinel 2 data. Three techniques will be compared to classify pixels: MNDWI, k-means and Random forest algorithm. When comparing performance using accuracy and kappa score, the best classification algorithm will be used to compare the extent of water throughout the data collection.

Figure 1: True color image of lake garda

1 Introduction & Classification

Lake Garda has an area of approximately 370 km^2 and it is located in the north of Italy, between Lombardia, Veneto and Trentino. It is surrounded by mountains, and this will create some additional cautions to evaluate cloud percentage and water extent. This will be explained better later.

The first step to evaluate water extent is to classify pixels as water or not. So the lake is isolated in the polygon called "lake" in the GEE project and all performances are evaluated on data elaborated on the first snapshot taken during spring-summer period of 2022. The True color image, included in this essay 1, shows morphological aspect described before and it is reconstruted using bands 2, 3 and 4. Let's get through each of the three classification algorithm(an example of classification can be seen on 3

1.1 MNDWI

Modified Normalized water index, as proposed in [Xu06] for a more accurate classification of rivers, lakes and open waters, is composed as follows:

$$\text{MNDWI} = \frac{Green - SWIR}{Green + SWIR}$$

It has been implemented as a first approach based on the spectral response, given a classification threshold of 0.3 for classifying water. 82 points were sampled(50 water, 32 not water) to test the classification:

	Not water	Water
Not water	32	0
Water	1	49

Metric	Value
Accuracy	0.9878
Kappa	0.9745

This type of classification is not well suited for a finite classification close to the lake's board, probably due to the lower level of the lake so with a "weaker" spectral response. However accuracy and kappa score are very high.

1.2 K-Means

The second approach is carried out by using an unsupervised clustering algorithm: K-means. Sampling randomly 10000 points into the lake polygon, it has been trained to identify only two classes (water vs not water). Results as follows:

	Not Water	Water
Not Water	27	5
Water	0	50

Metric	Value
Accuracy	0.9390
Kappa	0.8682

The accuracy has decreased with the classification of some not water points classified as water, usually near the board; this is the opposite situation of MNDWI classification. Let's discard this approach.

1.3 Random Forest

The third classification algorithm is a supervised one. Given previous problems, the accurate choice of training points is made by selecting points also close to the border to increase accuracy. The results are as follows:

	Not Water	Water
Not Water	32	0
Water	0	50

Metric	Value
Accuracy	1
Kappa	1

Here, the accuracy has increased a lot; also graphically, this algorithm is able to recognize the lake's board more precisely. However, a visual inspection of the classification on the map highlights that the algorithm still produces some misclassified pixels. Random forest classification will be used to analyze.

2 Comparison between 2022 and 2023

2.1 Cloud mask

One of the main problems when first applying random forest to the entire collection of data was the presence of clouds. These created some problems with the amount of pixel classified as water and produced some not reliable results. The first approach was filtering images using the "CLOUDY PIXEL PERCENTAGE" variable, selecting only images with low levels.

Two problems were present:

1. Some clouds were present on the lake surface altering the tot amount of water

2. The majority of cloudy pixels used to calculate percentage were not on the lake surface, but "above" mountains

Considering these, a more robust method has been implemented evaluating a cloud percentage only on the polygon "lake". Using values of "SCL" band that range from 1 to 11, in which bits from 8 to 10 are classified as clouds. So a real cloud percentage of clouds can be evaluated on the lake polygon. A threshold was set at only images with less than 1% of clouds.

Hence another small adjustment need to be done, because in the "clean" lake dates there is sometimes mist or fog which lead to misclassifications of pixels. To overcome this problem, a retrain of the random forest classifier also uses some points classified as water but on days with mist.

This created a fairly robust and reliable classification and data to be analyze.

2.2 Statistics and Results

The collection of images after all cleaning process were 27 in 2022 and 21 in 2023. These are some descriptive statistics about the two time series:

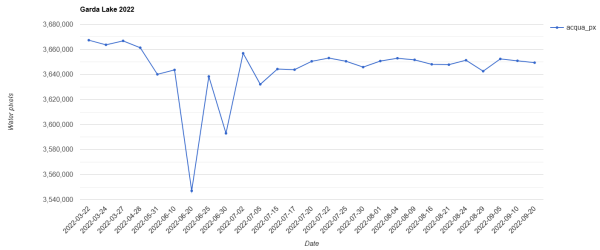
Table 1: Water pixel statistics (pixels and km^2) at 10 m resolution

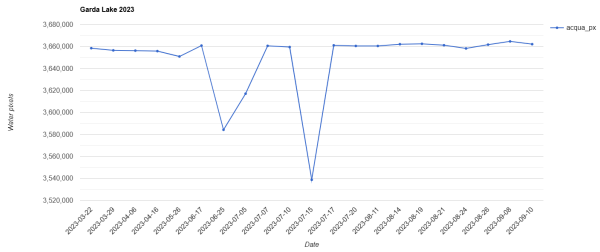
Year	Mean	Maximum	Minimum	Stand. Dev.
2022	3644302.92	3667350.41	3547023.57	23336.18
	364.43 km^2	366.74 km^2	354.70 km^2	2.33 km^2
2023	3648291.47	3664671.10	3538794.03	30519.92
	364.83 km^2	366.47 km^2	353.88 km^2	3.05 km^2

The difference between the two years, especially regarding mean is very similar: this could be a good result proving that the classification methodology was robust.

The lake area was slightly larger in 2023, in terms of mean and maximum values. However, the minimum is lower than the lowest area in 2022, proving that there was more variability on 2023, explaining the higher standard deviation (chosen wrt variance because it provides a directly applyable measure, in the same unit system).

Graphically the data collection, displayed as time series:





The first thing to point out is that dates on the horizontal axes are not the same, because they come from two cleaning process which does not guarantee the correspondence between two years(also the acquisition time is different). Secondly, notice also that the vertical axes does not start from zero, so the points which are from what could be a "median" observation , can be considered not a so relevant outlier (remember threshold for cloud coverage was 1%, so that variation could be a small cloudy area given the magnitude of the deviation).

Overall the values remained more stable around 3, 66 million water pixels in 2023, while 2022 was showing more variability. The maximum values were registered during april in 2022, while in 2023 between the end of summer. It can be interfered, with some grade of uncentarty, an opposite trend in 2022 with an overall decreasing comparing to a (small) increasing in water expansions.

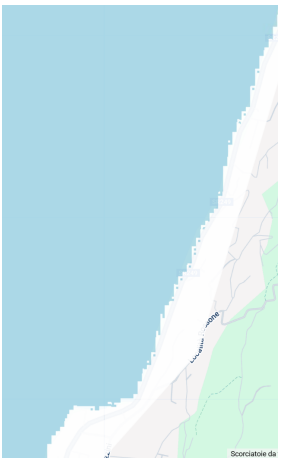


Figure 2: MNDWI classification

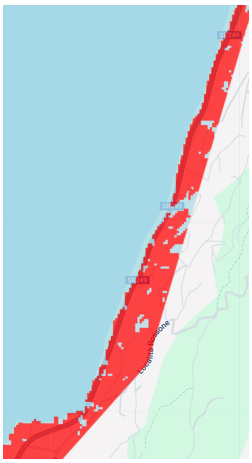


Figure 3: KNN classification

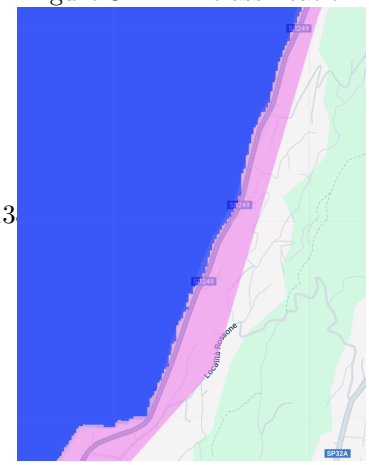


Figure 4: Random Forest classification

3 Code

The code can be found at this link: https://code.earthengine.google.com/21596e63ea8c77e5a4b8183e9d1e13

References

[Xu06] Hanqiu Xu. Modification of normalised difference water index (ndwi) to enhance open water features in remotely sensed imagery. *International Journal of Remote Sensing*, 27(14):3025–3033, 2006.